CSRMS

NADV 744 · Advanced Development Systems · Group Assignment 2026
Sol Plaatje University

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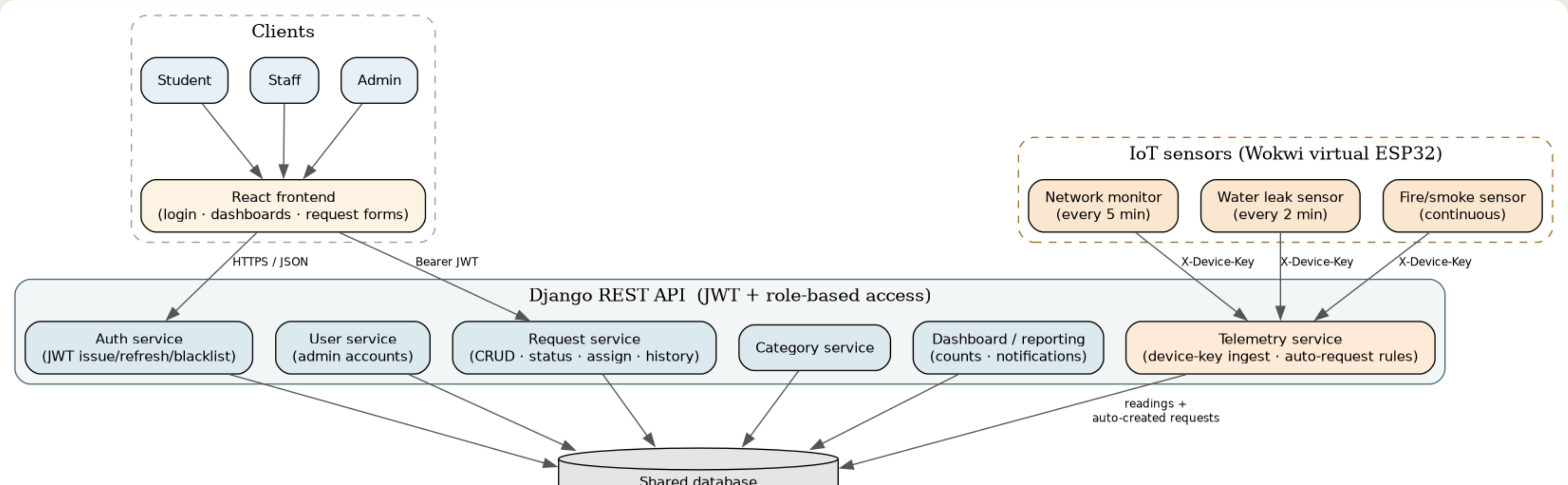
The problem

- Campus maintenance runs on WhatsApp and hallway conversations
- Reports arrive without structure: no category, no location, no owner
- Nobody can answer: *how many problems, how long to fix, which buildings?*
- Some failures **should not depend on a human noticing** — leaks and overheating
  happen at 2am

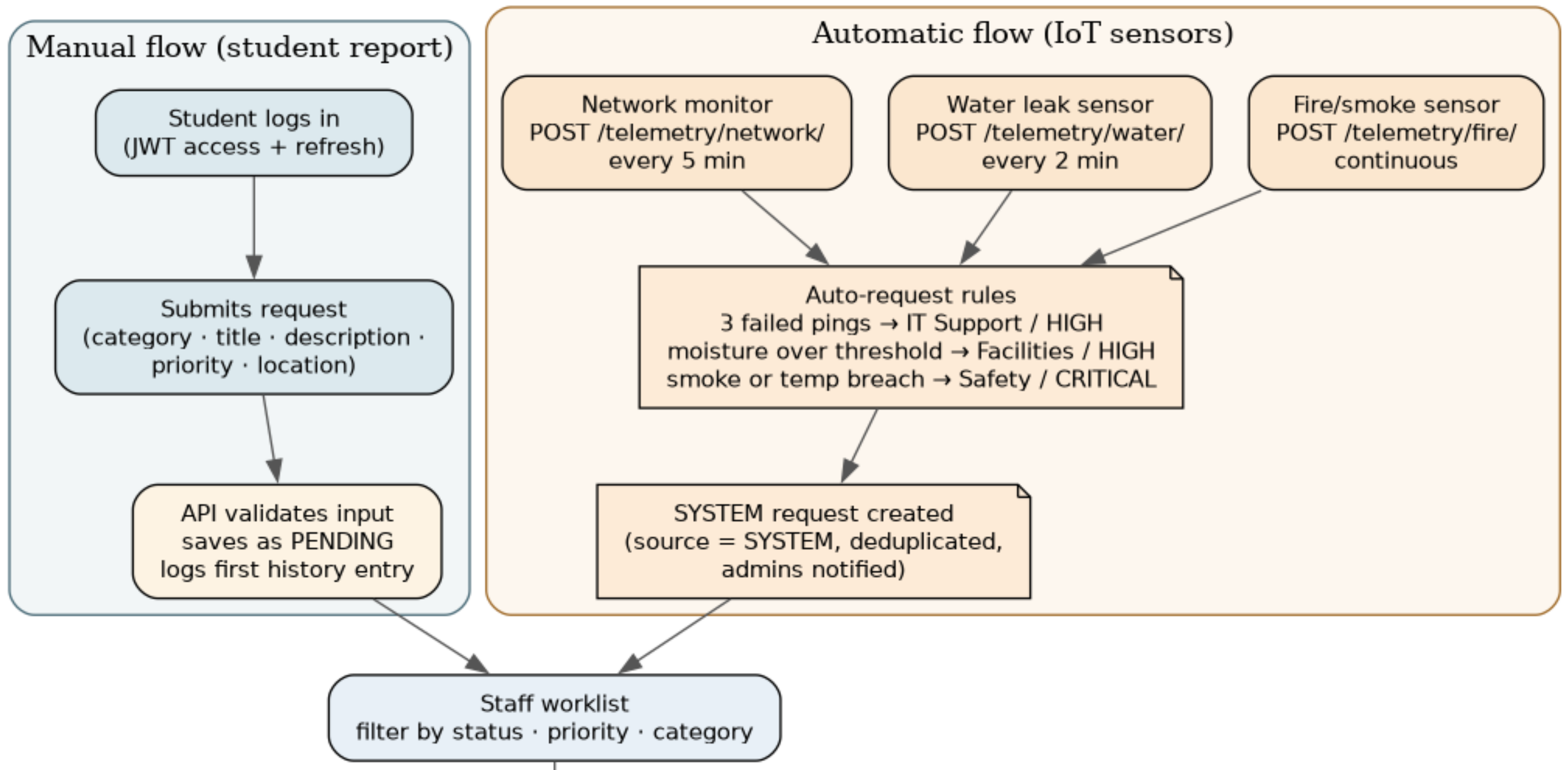
What we built

One system, two ways in:

- Students log problems → staff resolve them through a managed workflow
- Simulated IoT sensors raise requests **on their own** when conditions demand it



How a request travels



Security evidence

- Deliberate wrong-password login returns **HTTP 401** and grants no session
- A valid login returns JWT access and refresh tokens
- Students cannot read another student's request, even with a guessed ID
- A user JWT cannot post telemetry; sensors require a type-bound `X-Device-Key`
- Postman provides a visible API fallback for login, telemetry, and deduplication

Security model

Layer	Decision
Auth	JWT on every endpoint (register/login/refresh excepted)
Roles	STUDENT / STAFF / ADMIN — registration always creates STUDENT, server-side
Objects	A student who guesses another's request ID gets a **404**, proven by tests
Devices	Per-sensor keys, SHA-256 hashed, type-bound, individually revocable
Passwords	Django validators + PBKDF2

IoT auto-request rules

Sensor	Cadence	Trigger	Ticket raised
Network monitor	5 min	3 consecutive failed pings	IT Support · HIGH
Water leak	2 min	moisture > 60%	Facilities · HIGH
Fire/smoke	10 s	smoke ≥ 40 or temp ≥ 50 °C	Safety · CRITICAL

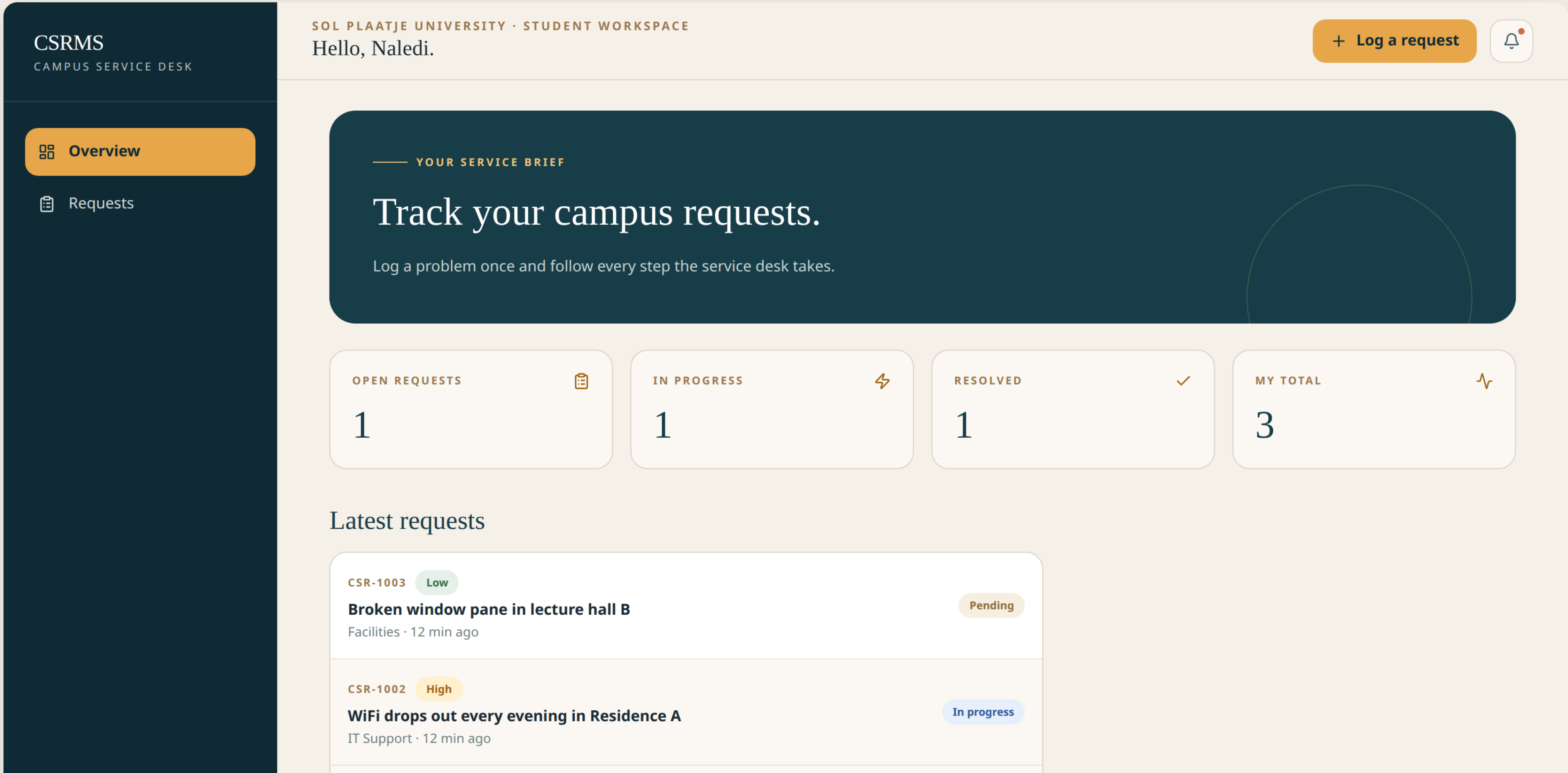
Open SYSTEM tickets are **deduplicated** per sensor+location — a stuck sensor raises one ticket, not a hundred.

Network, water, and fire/smoke triggers are verified by the backend tests and can be
demonstrated live or through Postman when Wokwi or the network is unavailable.

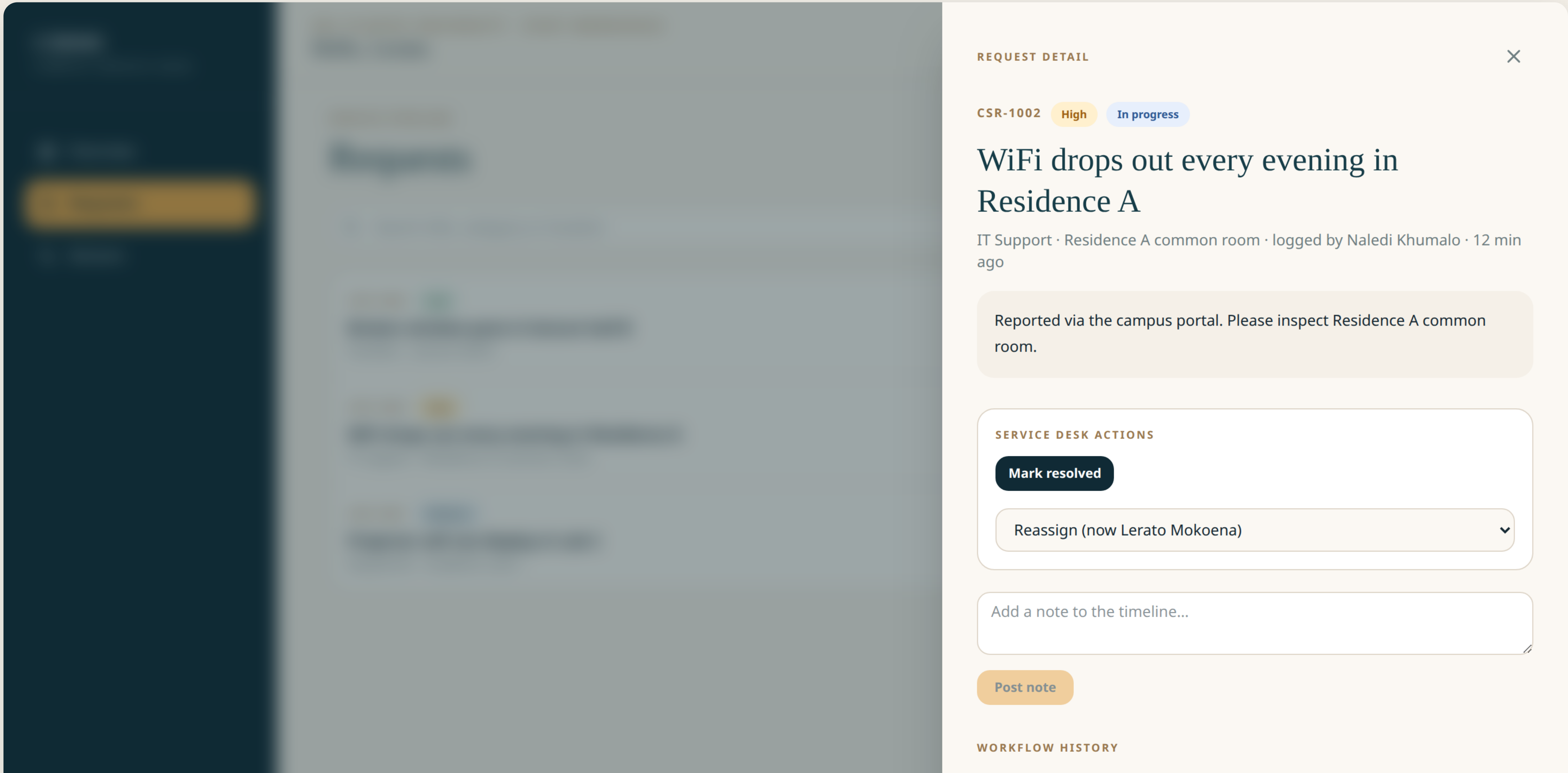
Live demo

Student report → staff workflow → sensor-triggered ticket

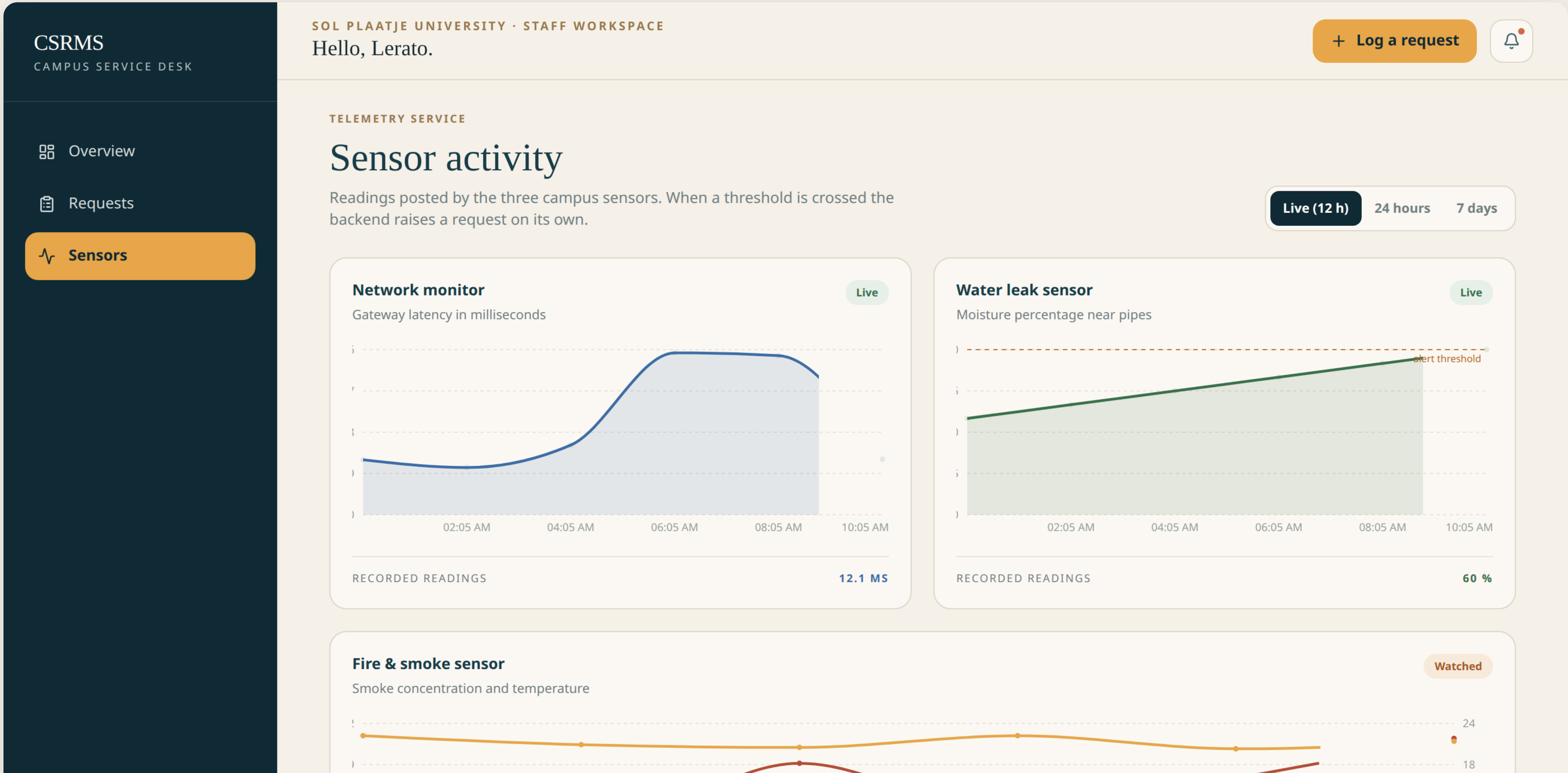
Student view



Staff workflow with full history



Live sensor telemetry



Testing & evaluation

- **61 automated API/service tests** — every security claim has a negative test
- Firmware: structural validation + Wokwi scenarios driving sensors to their limits
- Playwright walkthrough of all three roles doubles as the screenshot generator
- Postman collection covers the API as a manual demonstration and fallback
- Performance (50 rounds/endpoint): **p95 under 8 ms** on all reads; JWT checks are DB-free, lists are paginated

Limitations & future work

Honest limits: simulated hardware · pull-only notifications · laptop-scale perf data

Next, in order:

1. Email/push notifications with staff digests
2. SLA tracking and automatic escalation
3. Real ESP32 hardware (same sketches, new WiFi credentials)
4. MySQL deployment behind HTTPS

One workflow, recorded end to end

Thank you

github.com/altechemist/NADV74

`README.md` → running everything in three commands